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Palaeomagnetism of (late Vendian–earliest Cambrian) minor alkaline intrusions, Fen Complex, southeast Norway

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Emplacement of the Fen central complex (603–565 Ma) within the Fennoscandian Shield in southeast Norway was preceded by the emplacement of numerous minor alkaline intrusions into the surrounding gneisses. A palaeomagnetic sample of 28 of these bodies has identified a predominant SSE negative remanence carried by magnetite in some bodies and hematite in others. A sporadic high blocking temperature component appears to record localised effects associated with the development of the Oslo rift and igneous province to the east, but no major magnetic overprinting by post-emplacement events is recognised. The stable magnetisation vectors for twenty sites comprise a coherent population with those for two sites reversed with respect to the remainder; they yield a mean direction of $D = 210°$, $I = 44°$ ($\alpha_{95} = 6.4°$) and a palaeomagnetic pole at 324°E, 50°S ($dp/dm = 4.9°/7.9°$). The difference between the pole position for this early phase of the Fen magmatism and that for the late metasomatic rødberg (322°E, 63°S) in the interior of the complex is interpreted in terms of continental movement during the late Vendian–earliest Cambrian interval of alkaline activity here. The defined direction of APW movement continues a motion recognised from other Vendian data but subsequent movements during Lower Cambrian times are unclear.

1. Introduction

Several independent analyses of palaeomagnetic data have identified large apparent polar wander (APW) movements during Late Precambrian and Lower Cambrian times which imply high rates of continental movement during this interval. This is specifically seen in APW segments which contrast with the Upper Palaeozoic and younger paths (e.g. [1]) and the long latitudinal movement of the Gondwanaland path [2] which corresponds with the incidence of late Precambrian and Cambrian glaciation [3]. The scale of APW during these times is currently best defined from studies within the former supercontinent of Gondwanaland [3]; the definition and dating of these movements is, however, hampered by the paucity of igneous material of this age and most of the existing record comes from weakly deformed sedimentary platform successions (e.g. [2,4,5]) in which the age of the characteristic remanence is often unclear. Special interest has therefore focussed on the well-dated alkaline igneous complexes which are one facet of a major episode of continental rifting and break-up during these times [3,7]. Within the Fennoscandian Shield of the European block the Fen and Alnö bodies have both been studied [8–10], but the results have been generally disappointing and most of the facies within these central complexes appear to be poor recorders of the ancient geomagnetic field. However, emplacement of the Fen Complex was associated with, and preceded by, the intrusion of an intense swarm of small dykes and sheets of tinguaitic (porphyritic nepheline syenite) composition [11]. These bodies have proved to be more satisfactory recorders of a stable ancient remanence than the central complex and form the subject of the study reported here.

2. Geological outline

The Fen Complex is emplaced into Sveconorwegian (ca. 1100 Ma) gneisses in southeast Norway at 9°18′E, 59°18′N. It was apparently first emplaced as a liquid magma phase responsible for the alkali syenite (fenite) now preserved in the peripheral parts; the latter was subsequently modified by a range of hydrothermal and fluidised phases [12]. These phases were responsible for a

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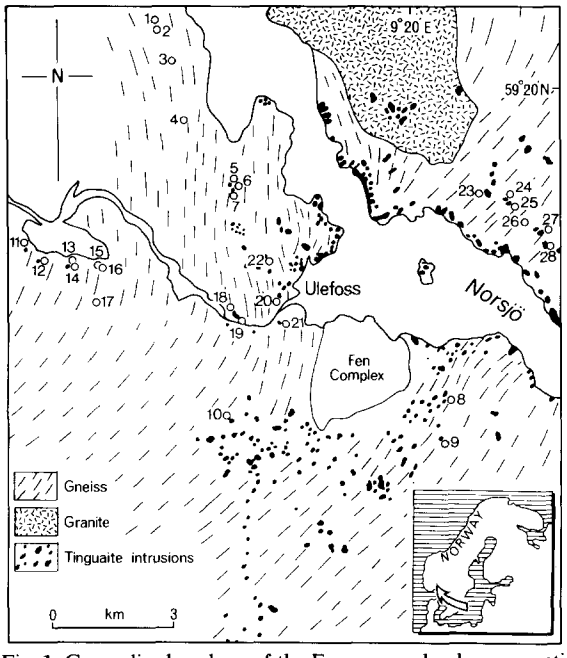


Fig. 1. Generalised geology of the Fen area and palaeomagnetic sampling sites. The distribution of tinguaite intrusions in the vicinity of the Fen Complex is taken from Bergstol [11], and the inset map shows the regional location of the sample area.

range of feldspar-free ultrabasic and carbonatitic rocks, including sövite and rødberg, which are now preserved in the central part of the complex, as well as kimberlite and kimberlitic breccias which occur as dykes and plugs respectively [12–14]. Radiometric (K-Ar biotite) ages of 599–572 Ma have been derived from the kimberlites [14,15] and other biotite ages from the sövites have yielded ages in the range 603–565 Ma [14,16]. The late phases of emplacement of this complex thus appear to have taken place at, or just before, the Vendian–Lower Cambrian boundary [17]. Alkaline magmatism commenced in this area with the emplacement of dyke-like and lensoid bodies of tinguaite ranging in thickness from 20 m down to a few centimetres [11]. The smaller bodies have low and variable angles of dip and comprise a dark fine-grained assemblage of alkali feldspar, nepheline, biotite and pyroxene [11]. They reach their greatest abundance within 3 km of the Fen complex (Fig. 1) but have been observed up to 25 km away. The dykes are cut by breccias recording an early phase in the emplacement of the central complex, as well as by carbonatites, kimberlite and dolerite dykes, and by the central complex itself.

3. Previous studies and the present sample

Poorter [8] investigated the palaeomagnetism of the Fen rødberg (a hematitic carbonatite) which represents part of the late metasomatic facies of the central complex containing up to 50% hematite. Mainly by applying partial alternating field (a.f.) demagnetisation, he isolated a remanence direction of $D = 205°$, $I = -56°$ from nineteen samples. Storetvedt [9] recognised a similar direction in four sites from the same facies. Although some samples retained this direction to 600 °C, others showed erratic movements above 500 °C, and the author preferred to regard this component as a resultant of shallow N- and S-directed components of inferred Upper Palaeozoic age; since no component analysis was undertaken, it was not possible to test this interpretation. Results from the kimberlites [9] identified a low magnetic stability with no clear groupings of directions. Two thoroughly altered dolerites from within the Fen complex (and therefore younger than the tinguaites) yielded Permian K/Ar ages; SE steep negative directions similar to those of the rødberg were recognised in the samples although some components were incompletely isolated at the limits of treatment [9, fig. 5].

Samples for the present study were collected from 28 sites within a radius of 16 km of the Fen Complex (Fig. 1). They come from fresh veins of tinguaite emplaced into country rock gneisses and are mainly exposed in road cuts on either side of Lake Norsjö. At each site between 5 and 11 independently drilled and oriented cores were collected. In the laboratory the samples were measured on sensitive parastatic magnetometers and were either progressively thermally treated in steps of 100 °C to 500 °C and then in reduced steps of 20 °C up to the Curie points of the magnetic carriers, or were otherwise demagnetised in a.f. in steps of 5–10 milli tesla (mT) to peak fields ranging from 40 to 140 mT. Components of magnetisation were visually determined from orthogonal vector projections and fitted by least squares; components were considered for further analysis if they were defined by three or more successive steps in the demagnetisation procedure.

4. Palaeomagnetic results

The tinguaites illustrate a wide range of responses to progressive demagnetisation, most typically between sites but sometimes within sites. The high coercivity samples are usually the most stable and define single components over most, or all, of the treatment (e.g. 13 in Fig. 2).

The lower coercivity material is typically somewhat less stable but a large soft component (presumably acquired in the Recent field) is always removed in the first two or three stages of treat-

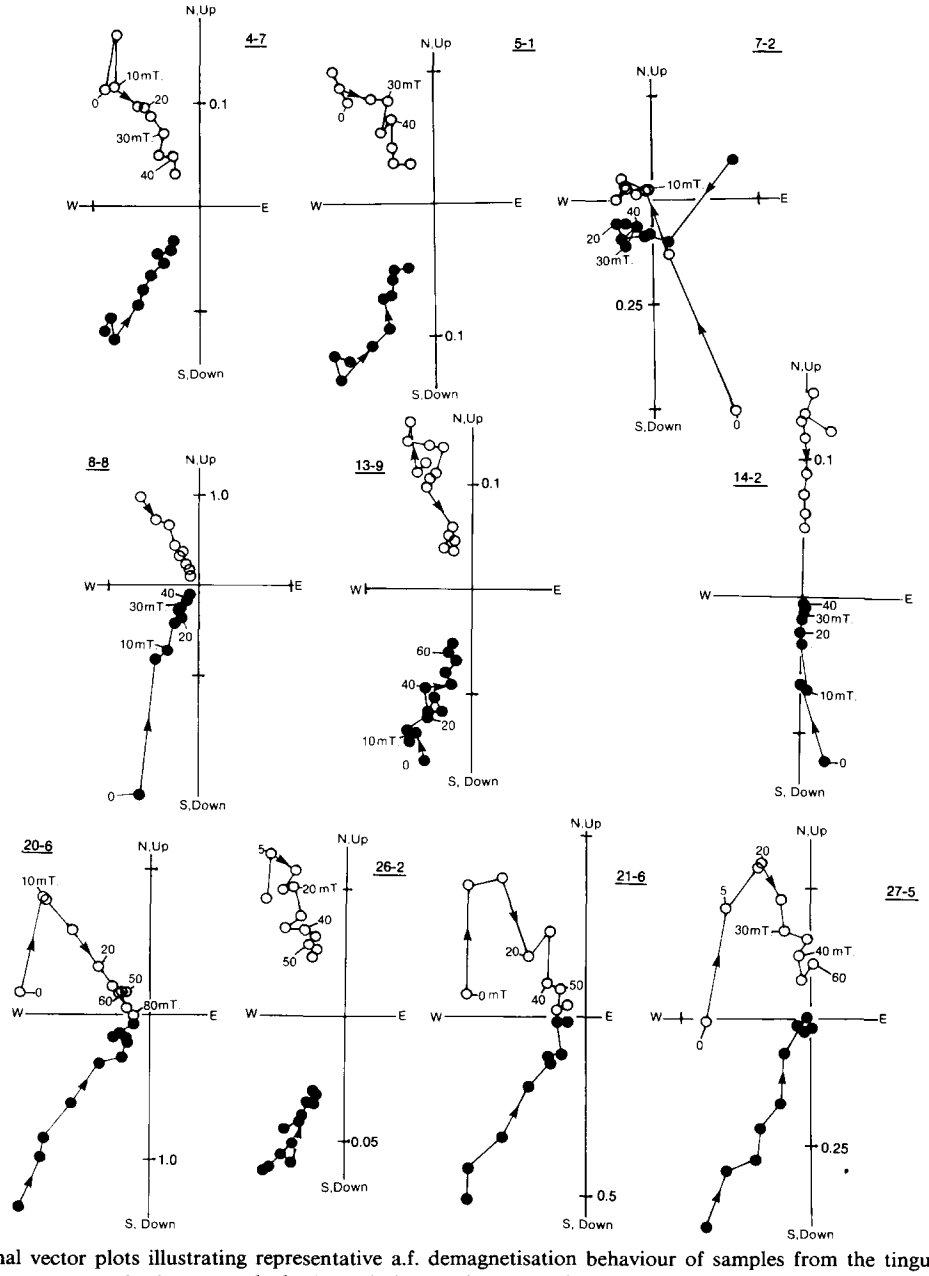


Fig. 2. Orthogonal vector plots illustrating representative a.f. demagnetisation behaviour of samples from the tinguaite collection. Closed symbols are vector projections onto the horizontal plane and open symbols are projections onto the vertical plane. Scale units of intensity are $\times 10^{-5}$ A m^2 kg^{-1}.

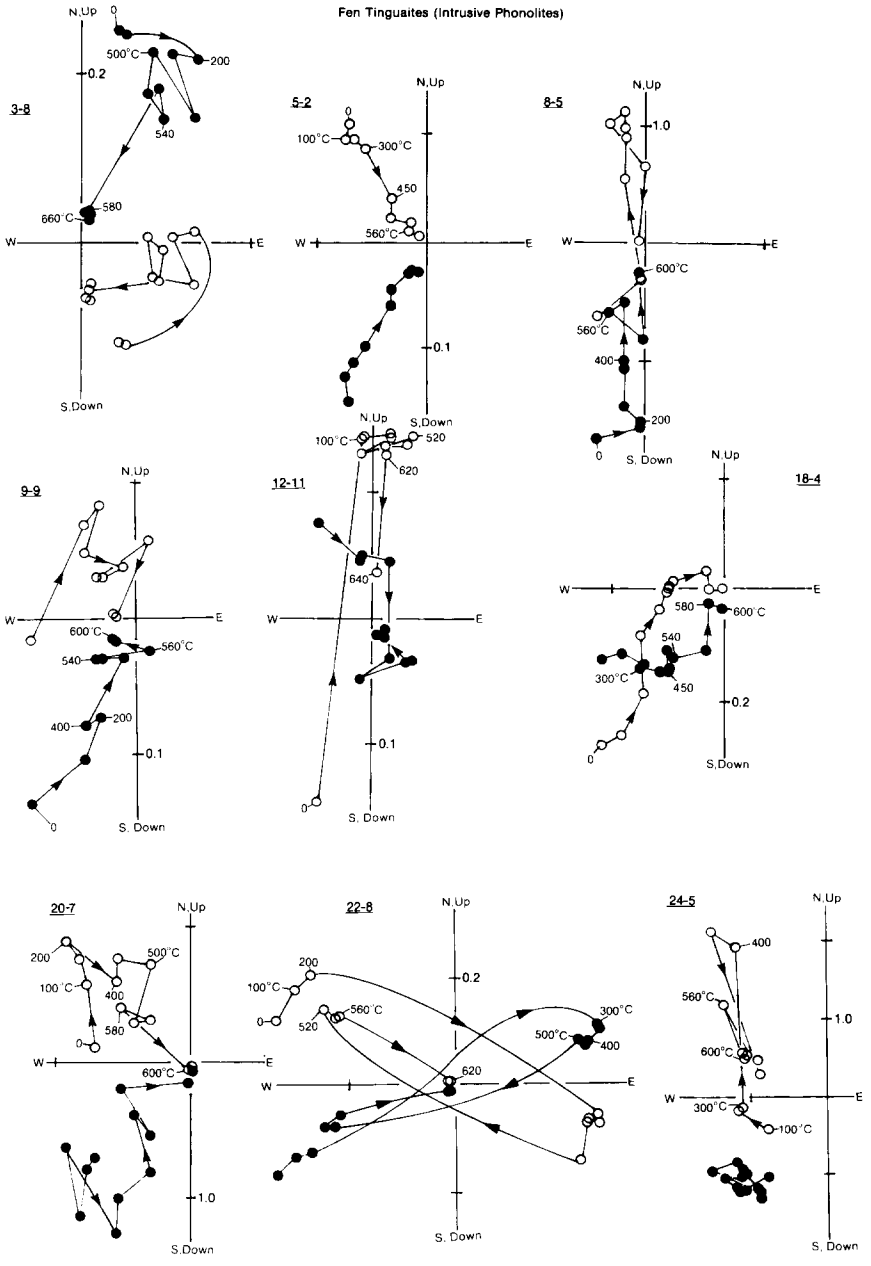


Fig. 3. Orthogonal vector plots illustrating representative thermal demagnetisation behaviour of tinguaite samples; symbols are as for Fig. 2.

ment (7, 8, 20 and 21 in Fig. 2). There is no systematic increase in the typical SSW negative inclination beyond these initial stages of a.f. treatment which could be interpreted as the subtraction of a shallow Permo-Carboniferous low coercivity overprint. Site 26 is typical of the high coercivities observed in the last batch of sites (22—27) where hematite carriers are present (24 in Fig. 3); the full vector cannot be recovered by a.f. cleaning in these samples although the migration on orthogonal plots is sometimes sufficient to demonstrate that the SSW negative direction is defined by a single component (Fig. 2). The SSW negative component (Fig. 4) is otherwise resolved

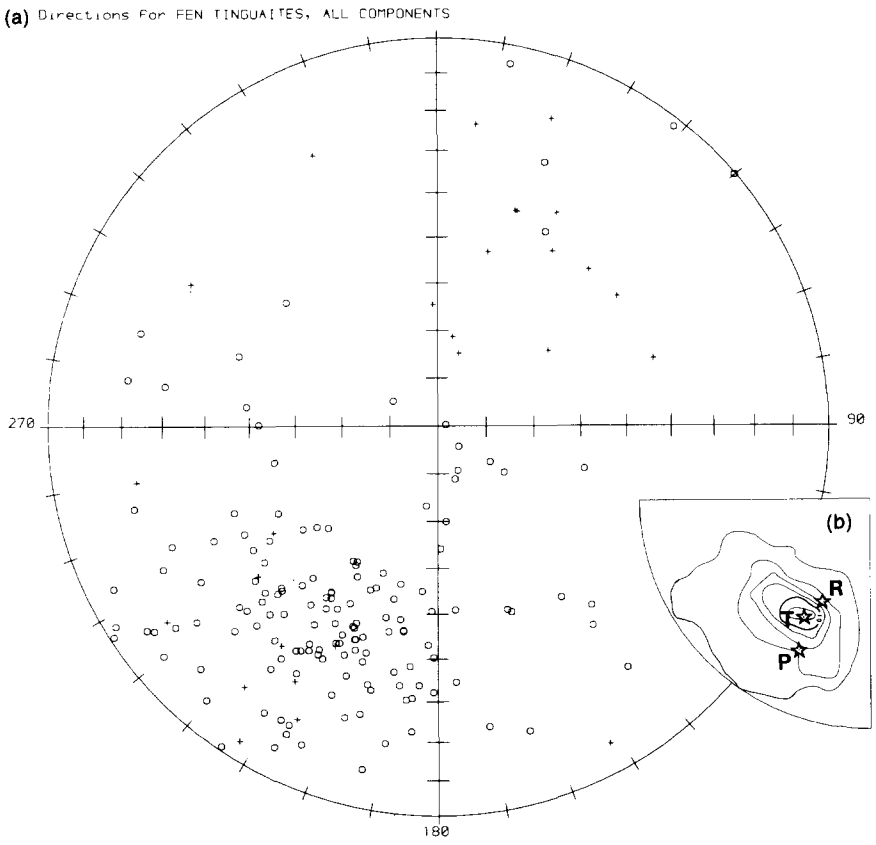


Fig. 4. (a) Stereographic projection illustrating directions of magnetisation of all components of magnetisation identified in the tinguaite collection as lower hemisphere (crosses) and upper hemisphere (open circles) plots. (b) SW quadrant of an equal area projection with the direction population of (a) plotted in the upper hemisphere (with the lower hemisphere directions reversed) and density contoured in divisions of 2%. The open stars are the mean directions derived from the rødberg (*R*), the tinguaites (*T*, calculated as noted in the text) and the mean of 15 Permo-Carboniferous studies from the Fennoscandian area (*P*). The rødberg is located in the interior of the Fen Complex (Fig. 1) and the Permo-Carboniferous results are derived mostly from the Oslo igneous complex immediately to the east of the sample area and the adjoining extension of the rift zone in southwest Sweden.

as a single magnetite-held remanence in many sites (e.g. 5 in Fig. 3).

Sites 3 and 28 yield a component close to antiparallel to this characteristic direction although in some samples (3 in Fig. 3) this is largely recovered in the higher part of the spectrum. As with a.f. cleaning, an increase in negative inclinations is only observed over initial stages (0–200 ° C) of thermal treatment in the majority of samples (8, 9 and 20 in Fig. 3). Whilst there is no significant evidence for a Permo-Carboniferous overprint in these rocks at the lower end of the blocking temperature and coercivity spectra, there is evidence in a minority of samples that a component with this direction is present at the highest end of these spectra. In samples 5 and 9 (Fig. 3) for example, the sample directions move back to shallower inclinations at the highest stages of treatment although these segments represent only about 10% of the predominant component. In some samples, such as 24 in Fig. 3, a movement towards shallower inclinations takes place in the higher stages of treatment but this high blocking temperature component is not fully resolved by treatment at temperatures close to the Curie point of hematite; a failure to subtract this component probably reflects the limitations of the thermal demagnetising apparatus. In one site (18, see Fig. 3) a SW shallow component is resolved by both thermal and a.f. cleaning in most samples and it seems likely that this unit has a Permo-Carboniferous remanence (and probably emplacement) age.

Two examples of non-typical behaviour are illustrated in Fig. 3: sample 12 shows a very steep negative component and could represent an overprint acquired late in the emplacement history of the central complex (see [8, fig. 1]) while sample 22 is a curious example of a double reversal of the characteristic remanence direction. This latter phenomenon has been observed during thermal cleaning of two samples below the Curie point of magnetite; its explanation is not clear but the resolved axis is somewhat shallower, and the inclination has a more easterly/westerly declination, than the main population.

The tinguaites are mostly fresh and fine-grained units which may be expected to have acquired an instantaneous record of the geomagnetic field during initial cooling. Hence the predominant components recognised in each sample have been combined at the site level to derive the data listed in Table 1. Although the veins have been locally shattered by younger brittle deformation associated with the emplacement of the central complex, no later regional deformation is apparent and the directions are therefore treated in situ. With the exception of two aberrant sites (sites 12, 18) excluded from the group mean calculated in Table 1, site 14 with a steep inclination, and site 15 with a more westerly declination, the remaining 18 sites show a somewhat higher distribution of declination (180° to 227°E) than inclination (−31° to −56°, Fig. 4). There is therefore no obvious case for regarding these data as smeared towards a

TABLE 1

Site mean palaeomagnetic results, minor intrusions bordering the Fen Complex, southeast Norway

Site No.	N (n)	D (°)	I (°)	R
3 [a]	10 (8)	20.7	54.4	7.73
4	8 (6)	204.4	−52.6	5.24
5	9 (8)	215.2	−37.6	7.84
7	6 (6)	211.3	−30.6	5.43
8	8 (7)	211.9	−43.2	6.65
9	9 (8)	214.8	−45.8	7.12
10	8 (6)	209.2	−44.8	5.45
11	10 (9)	198.4	−33.1	8.64
12 (i) [b]	11 (3)	33.4	−2.9	2.88
(ii) [b]	11 (3)	159.7	−85.2	2.95
13	11 (11)	204.3	−50.3	10.68
14	7 (6)	181.4	−71.7	5.75
15	8 (5)	241.9	−47.6	4.40
18 [b]	8 (7)	212.7	2.5	6.16
20	9 (8)	226.5	−25.6	7.66
21	10 (10)	215.4	−40.2	9.28
22	9 (8)	216.3	−40.1	7.63
23	8 (5)	179.2	−46.7	4.78
24	5 (4)	198.3	−36.0	3.88
25	10 (8)	194.6	−39.0	7.71
26	10 (8)	227.5	−22.8	7.11
27	8 (7)	198.7	−41.7	6.16
28 [a]	6 (6)	21.0	56.2	5.76

Group mean result

N	D (°)	I (°)	α_{95} (°)	R	Palaeomagnetic pole		
					°E	°N	dp/dm
20	210.4	−43.5	6.4	19.30	323.9	−50.1	4.9/7.9

D and I are the declination and inclination of the mean vector calculated from n samples out of N collected from the site. R is the resultant vector. Dp and dm are the semi-axes of the oval of 95% confidence about the pole position in the colatitude direction and at right angles to it respectively.

[a] Sites reversed for mean calcualtion.

[b] Sites excluded from group mean calculation.

younger ancient field direction; the inclination distribution of the original Fen rødberg data [8] interpreted by Storetvedt [9] as the result of smearing along a great circle passing through an Upper Palaeozoic field axis, is not apparent here.

5. Interpretation

Of major relevance to the interpretation of demagnetisation results from the Fen tinguaites is the possibility that a primary remanence acquired during igneous emplacement has been overprinted by the effects of one or more later tectono-magmatic events. This is especially so because one previous study has regarded the typical SSE negative component in the central complex as the resultant of N- and S-directed shallow components [9], and because this region is sited some 10 km from the western extremity of the Oslo rift which was the location of rifting and igneous activity during late Carboniferous and Permian times (and possibly during parts of Mesozoic times as well). Widely spaced dykes associated with the Permo-Carboniferous rifting are present throughout this area. The igneous rocks of the Oslo rift appear to have been emplaced or erupted mostly within the Permo-Carboniferous reversed superchron [17] and estimates of the ancient field direction relative to southeast Norway are ca. $D/I = 200°/-30°$ [9,18,19]. The Fen rocks were otherwise emplaced within an already cooled and stabilised Precambrian shield removed by more than 150 km from the Caledonian orogenic belt to the west, and from the sites of the late Mesozoic and younger rifting to the south and west.

The group mean direction derived from the tinguaites ($D/I = 210°/-44°$) has a similar declination, and is somewhat shallower in inclination, to the mean direction derived from the rødberg ($D/I = 205°/-56°$ [8]). Although this difference might be regarded as the result of contamination of the tinguaite remanence by a small component of a reversed Permo-Carboniferous overprint compounded with the late metasomatic remanence in the Fen Complex, this explanation is discounted for five reasons: (1) there is no systematic movement towards, or away from, a reversed shallow direction during progressive cleaning in the bulk of the collection; (2) with the exception of two examples noted by Storetvedt [9], the K/Ar systems have not been reset by the Permo-Carboniferous activity in this area [15]; (3) tinguaites are mainly a fresh chilled facies unlike the metasomatically-altered facies from the interior of the complex where even a dolerite yielding Permian K/Ar ages does not yield a typical Permian field direction [9]; (4) a reversal of ancient field polarity is defined by some sites, and (5) palaeomagnetic studies of the basement terrane in this area demonstrate that it preserves a Precambrian uplift and cooling remanence acquired following the Sveconorwegian mobile episode (ca. 1050–850 Ma) and it does not appear to be significantly overprinted by the Permo-Carboniferous tectono-magmatic activity associated with the Oslo rift ([20] and unpublished results of the author). However, a weak overprinting in the higher blocking temperature fraction of the remanence spectrum has been identified in some sam-

TABLE 2

Late Precambrian–Lower Cambrian palaeomagnetic results from the Fennoscandian Shield

Rock unit	Age	Pole position	*dp/dm*	Palaeolatitude	Reference
1. Hunnedalen dykes	< 950–850 RS, KA	28° E, 34° N	8/9	62	[21]
2. Egersund dolerites (i)	< 950–850 RS, 850–663 KA	50.8° E, 22.4° N	14/16	42	[21]
(ii)		52.0° E, 28.0° N	15/17	55	[22]
3. Batsfjord dykes	640 KA	108.2° E, 40.6° N	5.1/6.7	43	[23]
4. Fen tinguaites	> 603–565 KA	143.9° E, 50.1° N	4.9/7.9	25	this study
5. Fen rødberg	603–565 KA	142° E, 63° N	3/4	37	[8]
6. Nexo sandstone	Lower Cambrian	134° E, 38° N	7/11	16	[24]
7. Alnö complex (i)	589–544 KA, 553 RS, 554 KA	300° E, 85° N	19/20	63	[10]
8. Alnö complex (ii)		92° E, 8° S	5/9	3	[10]

KA = potassium-argon, RS = rubidium-strontium. *Dp* and *dm* are the radius of the oval of confidence about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively. The palaeolatitudes are calculated from the ancient magnetic inclinations in the study areas which range from the Varanger Peninsula in the north to the Island of Bornholm in the south.

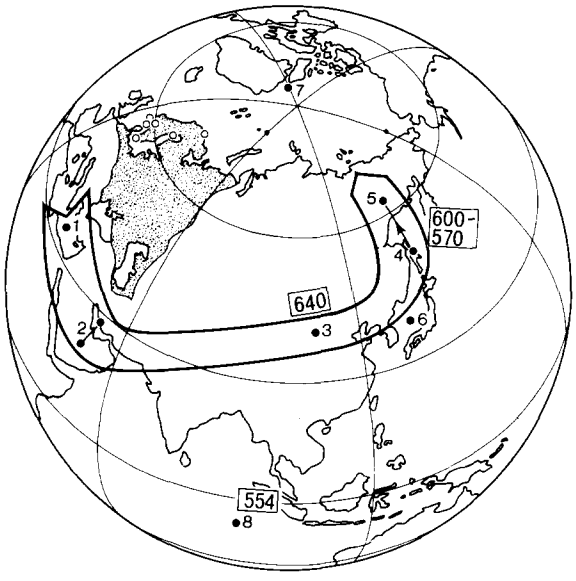


Fig. 5. Vendian to early Cambrian apparent polar wander path for the Fennoscandian Block derived from the palaeomagnetic poles listed in Table 2. This tectonic division is generally regarded as integral with the basement extending to the Uralian suture beneath the Russian platform as shaded in this figure; the locations of the studied rock units yielding the palaeomagnetic poles are indicated by the open circles. As discussed in the text, the continuity of the path beyond the Fen poles (*4* and *5*) is uncertain.

ples from this study; this would be compatible with a patchy oxidisation of these rocks by circulating fluids during the Permo-Carboniferous activity, but it has evidently not had a major influence on the remanence record in these rocks.

For the above reasons the difference in magnetic inclination between the tinguaites and the rødberg is interpreted here in terms of a movement of the European block into higher latitudes (Table 2) during the latest Precambrian to early Cambrian interval recorded by the Fen alkaline magmatism. The collective results thus define the direction of APW movement during these times (Fig. 5). Unfortunately Vendian to Cambrian data from this crustal block are not yet adequate to properly define the preceding and succeeding paths connecting with this segment. The Fen results appear to continue a movement defined by results from dyke swarms (results 1, 2 and 3 in Table 2 and Fig. 5) recording episodes of late Precambrian fracturing within the Fennoscandian Shield. The path may then extend to a result from a steep component in the ca. 40 Ma younger Alnö Complex (pole 7); however, this latter pole is computed from a steep positive remanence which may be dominated by younger overprints. The predominant dual polarity axis in the Alnö body (pole 8) has a shallow inclination and a return of this block to low palaeolatitudes during Lower Cambrian times is also suggested by a result from the Nexo Sandstone (pole 6). Many more studies will be required to evaluate the detail of this APW path and they will, unfortunately, have to focus on sedimentary material.

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